

Department Overview Department Name: Department of Information Technology Faculty: Faculty of Information & Communication Technology (FICT) Accreditation: The program is officially accredited by the National Computing Education Accreditation Council (NCEAC). National Standing: Ranked by the Higher Education Commission (HEC) as one of the top undergraduate programs in the field of IT in the country. History: Founded in 2007 as an undergraduate program, the department later expanded its offerings to include an MS in Information Technology. It features a vibrant, research-oriented faculty. Core Focus: Programming, systems analysis and design, business telecommunications, database management systems, computer communication, networking, infrastructure, information security, software development, artificial intelligence, enterprise systems, Internet of Things (IoT), and technology implementation management.

1. Bachelor of Science in Information Technology (BSIT)

Admission Requirements Academic Background: F.Sc / ICS with Mathematics from any recognized board (or equivalent qualification) with at least 50% marks. Alternatively, a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of 60% marks.

Degree Requirements Total Credit Hours: 133-134 • Total Courses: 37-43 Minimum Academic Standing: CGPA $\geq$ 2.0

Curriculum Plan (Program Schema by Semester) Note: Credit hour listings follow the format (Theory + Lab).

1st Semester • IT102/IT102L | Introduction to ICT | Prerequisite: None | Credit Hours: 2+1 • CS-114/CS-114L | Programming Fundamentals | Prerequisite: None | Credit Hours: 3+1 • HUM-364 | English Composition & Comprehension | Prerequisite: None | Credit Hours: 3+0 • MATHP-105 | Basic Mathematics | Prerequisite: None | Credit Hours: 3+0 • PHY-205/PHY-205L | Applied Physics | Prerequisite: None | Credit Hours: 2+1 • HUM-101 | Islamic Studies | Prerequisite: None | Credit Hours: 2+0

2nd Semester • CS-212/L | Object Oriented Programming | Prerequisite: Programming Fundamentals | Credit Hours: 3+1 • ENG-361 | Communication & Presentation Skills | Prerequisite: None | Credit Hours: 3+0 • MATHP-118 | Calculus & Analytical Geometry | Prerequisite: Basic Mathematics | Credit Hours: 3+0 • HUM-102 | Pakistan Studies | Prerequisite: None | Credit Hours: 2+0 • TBA | University Elective - I | Prerequisite: None | Credit Hours: 3+0 • MATHA-234 | Discrete Structures | Prerequisite: None | Credit Hours: 3+0

3rd Semester • CS-214/L | Data Structure & Algorithms | Prerequisite: Programming Fundamentals | Credit Hours: 3+1 • CS-332/L | Database Systems | Prerequisite: Introduction to ICT | Credit Hours: 3+1 • TE-307/L | Computer Networks | Prerequisite: None | Credit Hours: 3+1 • TBA | University Elective II | Prerequisite: None | Credit Hours: 3+0 • MATHP-111 | Linear Algebra | Prerequisite: None | Credit Hours: 3+0

4th Semester • CS-341/L | Operating Systems | Prerequisite: None | Credit Hours: 3+1 • IT-332/L | Web Technologies | Prerequisite: Programming Fundamentals | Credit Hours: 2+1 • IT-321 | Information Security | Prerequisite: None | Credit Hours: 3+0 • STAT-101 | Probability & Statistics | Prerequisite: None | Credit Hours: 3+0 • TBA | University Elective III | Prerequisite: None | Credit Hours: 3+0

5th Semester • HUM-266 | Technical & Business Writing | Prerequisite: None | Credit Hours: 3+0 • IT-421/L | System & Network Administration | Prerequisite: Computer Networks | Credit Hours:

3+1 • CE-303 | Software Engineering | Prerequisite: None | Credit Hours: 3+0 • IT-341 | IT Infrastructure | Prerequisite: None | Credit Hours: 3+0 • CS-231/L | DB Administration and Management | Prerequisite: Database Systems | Credit Hours: 3+1

6th Semester • TBA | IT Supporting I | Prerequisite: None | Credit Hours: Not listed • TBA | IT Elective I | Prerequisite: None | Credit Hours: Not listed • IT-303/L | Virtual Systems & Services | Prerequisite: Operating Systems | Credit Hours: 3+1 • MGMT-212 | IT Project Management | Prerequisite: None | Credit Hours: Not listed • TBA | IT Elective II | Prerequisite: None | Credit Hours: Not listed

7th Semester • TBA | Final Year Project-I | Prerequisite: None | Credit Hours: 0+3 • TBA | IT Elective III | Prerequisite: None | Credit Hours: 3+0 • CE-132 | Cyber Security | Prerequisite: Information Security | Credit Hours: 3+0 • TBA | IT Supporting II | Prerequisite: None | Credit Hours: 3+0 • TBA | IT Supporting III | Prerequisite: None | Credit Hours: 3+0

8th Semester • TBA | Final Year Project-II | Prerequisite: None | Credit Hours: 0+3 • TBA | University Elective IV | Prerequisite: None | Credit Hours: 3+0 • TBA | IT Elective IV | Prerequisite: None | Credit Hours: 3+0 • TBA | IT Elective V | Prerequisite: None | Credit Hours: 3+0 • HUM-302 | Professional Practices | Prerequisite: None | Credit Hours: 3+0

2. Master of Science in Information Technology (MSIT)

Admission Requirements • Academic Background: 16 years of education or an equivalent qualification (such as a 4-year BS or B.E. degree) in a relevant discipline from an HEC-recognized university. • Minimum Marks: At least 60% marks under the annual grading system, or a minimum CGPA of 2.5 out of 4.0. • Testing: Candidates must pass an entrance test administered by the university or its authorized testing agency with a minimum cumulative score of 50%.

Degree Requirements Total Credit Hours: 30 Course Distribution: 8 to 10 graduate courses, consisting of 24 credit hours of graduate-level coursework and 6 credit hours of research thesis. Minimum Academic Standing: CGPA $\geq$ 2.5 Research Track Notice: Thesis research is conducted under the guidance of PhD faculty members. (Note: The page layout text containing a placeholder description errantly references "telecommunication systems engineering and power systems engineering" under the general MS degree framework).